

paper_04_agent_spacetime_architecture

Paper 4 Brief: Embedding Autonomous Agents in Simulated Spacetime

Authority state: INTERNAL_RESEARCH

Claim boundary: repo_verified_architecture; sensitive security framing still requires IP/release review.

Abstract

This paper converts the organism-agent and battleground material in the canonical packet into a bounded architecture preprint. It studies the design pattern of embedding autonomous agents and operational zones inside a simulated spacetime so that routing, isolation, influence, communication, and quarantine can be represented spatially. Because this thread contains security-oriented tiering, adversary-lab semantics, and potentially strategic routing mechanisms, publication should remain high-level unless Medina approves the release boundary. The paper should present the architecture as a design proposal and experimental framework, not as a verified production security system.

Distinct Thesis

Spatial embedding offers a design language for autonomous-agent coordination in which operational roles, trust boundaries, influence, and quarantine can be modeled as geometric regions and physical interaction rules.

Contribution Surface

- Three-tier signal classification: cooperative, hostile, shadow.
- Agent placement as position-bearing entities in a simulated substrate.
- Zone definitions as bounded geometric regions with semantic roles.
- Influence and communication modeled through force analogies.
- Architecture bridge from physical substrate to operational intelligence routing.

Bounded Claims

Claim	Class	Proof posture	Public wording
The architecture embeds agents as physical entities in simulated 3D space.	C5 / partial C1	Repo-backed through simulation/engine.py, simulation/world.py, and simulation/routes.py; full 21-agent physical placement not all visible in resident simulation list.	"The implementation embeds resident agents, threats, signals, honeypots, data packets, and defensive nodes in a tick-based 3D operational space."
Three-tier routing separates cooperative, hostile, and unknown signals.	C5 / C8	Repo-backed as README architecture; release-sensitive.	"The design documents a tiered routing model for cooperative, adversarial, and unknown inputs."
Gravitational attraction models influence or authority.	C4 / C9	Metaphorical/protocol claim; not empirical proof.	"The architecture uses physical analogies, such as attraction, to model influence relationships."
The system is production-oriented.	Potential C1	Repo-backed deployment scaffolding; live production operation not proven.	"The repository contains production-oriented deployment scaffolding, including Docker, compose, and CI build/test workflow."

Proposed Outline

1. Introduction: agent systems as spatial architectures.
2. Separation between physics substrate and operational semantics.
3. Signal classification and routing model.
4. Agent embedding: positions, zones, and interaction surfaces.
5. Zone geometry as policy boundary.
6. Influence, communication, quarantine, and isolation as spatial operations.
7. Release-safe architectural diagrams.
8. Evaluation plan: routing correctness, containment, latency, and failure modes.
9. IP-sensitive details and public/private boundary.

Required Evidence Before Preprint Release

- Agent registry or architecture diagram approved for release.
- Current repo-backed references: README.md, spatium-computationis/main.py, spatium-computationis/simulation/engine.py, spatium-computationis/simulation/world.py, spatium-computationis/defense/honeypot/routes.py, spatium-computationis/platform_router.py, Dockerfile, docker-compose.yml, and .github/workflows/deploy-app.yml.
- Non-sensitive pseudocode for tier classification and routing.
- Simulation traces showing movement, routing, quarantine, or zone transitions.
- Threat-model document and containment criteria.

- IP review deciding which names, zones, coordinates, and mechanisms may be public.

Minimized Overlap Rule

This paper should cite Papers 1-3 as substrate layers and focus on operational semantics. It should not restate the mathematical derivations or cosmology machinery except to explain how they constrain the agent architecture.